A

Major Project

On

NEURAL NETWORK-BASED PREDICTIVE MODELING FOR COMPLEX PSYCHIATRIC DISORDER IDENTIFICATION USING EEG FEATURES

(Submitted in partial fulfillment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

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CERTIFICATE

This is to certify that the project entitled "**NEURAL NETWORK-BASED PREDICTIVE MODELING FOR COMPLEX PSYCHIATRIC DISORDER IDENTIFICATION USING EEG FEATURES**" being submitted by **G.Tejaswini (227R1A1292),M.RitheshPavan(227R1A12B1)&A.HariVardhan (227R1A1267) ,** in partial fulfillment of the requirements for the award of the degree of B.Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

The early detection of neurological and psychiatric disorders is critical for optimizing patient outcomes and improving the efficacy of healthcare delivery. This study presents a novel multiclass machine learning (ML) framework designed to classify epilepsy, migraine, and schizophrenia simultaneously using electroencephalography (EEG) signals. Mental task identification using electroencephalography (EEG) signals is required for patients with limited or no motor movements. A subject-independent mental task classification framework can be applied to identify the mental task of a subject with no available training statistics. Deep learning frameworks are popular among researchers for analyzing both spatial and time series data, making them well-suited for classifying EEG signals. METHOD. In this paper, a deep neural network model is proposed for mental task classification for an imagined task from EEG signal data. The system implements a robust web application built with Flask that allows for user management, role-based access control (Admin for training/analysis and User for prediction), and a clear interface for analysis. The core methodology involves training and comparing multiple machine learning models including Random Forest, Support Vector Machine, Logistic Regression, and Decision Tree and a specialized Hybrid Graph Neural Network (HGNN) architecture. These models are trained to classify both main and specific psychiatric disorders based on processed EEG feature data. The objective is to evaluate the predictive performance of these models, particularly the HGNN's ability to capture complex non-linear relationships in brain connectivity data, providing clinicians and researchers with a high-accuracy, deployable tool for potential aiding in the preliminary diagnosis of psychiatric conditions.

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1.INTRODUCTION

1. INTRODUCTION

Neuropsychiatric and neurological conditions, particularly schizophrenia, epilepsy, and migraine, constitute a substantial global disease burden, with profound implications for individual wellbeing and healthcare infrastructure worldwide. Current epidemiological studies reveal striking prevalence figures, with schizophrenia affecting about 1% of the global population, migraine disorders impacting nearly 1 billion individuals, and epilepsy exhibiting a lifetime prevalence of 0.5–1%.

These statistics highlight the critical necessity for developing precise and accessible diagnostic methodologies. Modern neurodiagnostic techniques, including structural MRI, fMRI, and PET, are valuable in clinical settings but face challenges such as expense, limited availability in low-resource areas, and suboptimal temporal resolution. In comparison, electroencephalography (EEG) provides distinct benefits, being non-invasive, cost-effective, mobile, and capable of capturing neural activity with millisecond precision such as qualities that enhance its utility for studying cortical activity in neuropsychiatric disorders

1.1 PROJECT PURPOSE

The purpose of this project is to develop an intelligent system that can automatically identify psychiatric disorders using EEG (Electroencephalography) signals and machine learning techniques. Traditional diagnosis of neurological and psychiatric disorders often relies on manual interpretation of EEG signals by medical experts, which can be time-consuming, subjective, and prone to inconsistencies.

This project aims to overcome these limitations by creating a data-driven predictive framework that analyzes EEG features and classifies psychiatric disorders accurately. By integrating multiple machine learning models such as Random Forest, Support Vector Machine, Logistic Regression, Decision Tree, and a Hybrid Graph Neural Network (HGNN), the system can learn complex patterns in brain signal data and improve diagnostic reliability.Another key purpose of the project is to provide a web-based platform using Flask that allows administrators to train and evaluate machine learning models while enabling users to upload EEG feature datasets and obtain real-time disorder predictions.

1.2 PROJECT FEATURES

The proposed system provides an intelligent platform for identifying psychiatric disorders using EEG signal features and machine learning techniques. It integrates multiple classification algorithms such as Random Forest, Support Vector Machine, Logistic Regression, Decision Tree, and Hybrid Graph Neural Network to analyze complex brain signal patterns and improve prediction accuracy. The system includes a robust data preprocessing module that handles missing values, noise removal, feature selection, and normalization to ensure high-quality input data for model training.

Another key feature is the **Flask-based web application**, which provides a user-friendly interface for both administrators and regular users. Administrators can upload datasets, perform exploratory data analysis, train models, and compare classifier performance using evaluation metrics such as accuracy, precision, recall, and F1-score. Regular users can upload EEG feature datasets in CSV format and receive real-time predictions for psychiatric disorders. The system also supports model storage and reuse, allowing trained models to be saved and loaded efficiently without retraining. Overall, the platform provides an automated, scalable, and reliable solution for EEG-based psychiatric disorder classification and decision support.

2.LITERATURE SURVEY

2. LITERATURE SURVEY

Li et al. [1] established a robust framework combining empirical mode decomposition (EMD) with support vector machines (SVMs), achieving exceptional performance metrics (98.00% sensitivity, 99.40% specificity) through careful analysis of intrinsic mode functions (IMFs) and their statistical properties. Subsequent research by Satapathy et al. [2] demonstrated the effectiveness of transformer architectures in this domain, reporting 99.5% classification accuracy when processing wavelet-transformed EEG data. Manali et al. [3] have proposed a mental task classification using variational mode decomposition (VMD) to extract features from the single-channel EEG. There were three stages of processing in their work.

They first decomposed the signal using VMD and then calculated the variational mode energy ratio proposed in their work followed by an adaptive boosting algorithm for the classification purpose. Feature reduction is a crucial step in any machine learning task and has been studied by Conrado et al. [4] for the classification of mental tasks using ANNs. The convolutional neural network (CNN) has also been widely used by many researchers. In their work, Pallavi et al. [5] studied the image processing capability of a CNN. They used the scalogram images of EEG data for the classification of different emotions. The model developed by these authors was tested for different datasets and was found to be subject-independent. The Bidirectional Long Short-Term Memory Network (BiLSTM) proposed by Jinru et al. [6] was also used for the classification of various emotions using EEG signals. EEGNet is another CNN-based model developed for EEG-based BCIs. In their work, the authors used depth-wise and separable convolutions for model development. They compared the results obtained for cross-subject and within-subject classifications with the different approaches across the four BCI paradigms, namely, P300 visual-evoked potentials, ERN, MRCP, and SMR. Madhuri et al. [7] classified hand movement and word generation using a Hierarchical classifier that employed optimized Neural Networks on the EEG signals. In 2014, Xiu et al. [8] applied the DL algorithm for the classification of EEG data extracted for the Motor Imagery task (MIT). The two tasks studied by these researchers were the imagination of the left hand and the right hand motor activities. Saadat et al. [9] used Back Propagation Neural Network (BPNN) along with the Hidden Markov Model (HMM) for the classification of mental tasks. In their work, Jose et al. [10] studied various online learning mechanisms used in brain–computer interfaces (BCI) that can help in obtaining fixed learning rates in patients with neural disorders.

Debarshi et al. [11] studied subject-independent and subject-dependent models separately for EEG-based emotion detection and classification. From this work, they concluded that conventional machine learning techniques work better in the case of subject-independent decision making.

Chen et al. [12] conducted a systematic evaluation of six ML classifiers, identifying least squares

further methodological refinement. Yin et al. [13] developed a decision support system for clinical migraine diagnosis, focusing on differentiating migraine from tension-type headache. Their approach used expert-selected clinical features with K-nearest neighbors (KNN) classification, achieving 90% accuracy.

In EEG-based studies, Akben et al. [14] applied photic stimulation at 2 Hz, 4 Hz, and 6 Hz frequencies, analyzing power spectrum characteristics with ANN classification to reach 85% accuracy. Further advancing this approach, Subasi et al. [15] employed DWT for feature extraction from EEG signals during photic stimulation, using random forest (RF) classification to achieve 85.95% accuracy. For migraine subtype differentiation, Frid et al. [16] implemented a radial basis function SVM classifier that distinguished migraine with aura from that without aura with 84.62% accuracy .

For example, Movahed et al. [17] designed a classification framework for depression by extracting the statistical, frequency domain, and brain region association features of EEG, which can accomplish a classification accuracy of 99%. Qiao et al. [18] introduced a TanhReLU-based Convolutional Neural Network (CNN) for EEG-based classification of Major Depression Disorder (MMD), offering a promising accuracy of 98.59%. Gupta et al. [19] developed a privacy-preserving federated learning-based multimodal system utilizing Bidirectional Long Short-term Memory (BiLSTM) through audio and EEG data, reaching an accuracy of 99.9% for the MMD detection task. Hu et al. [20] presented an Iterative Gated Graph Convolutional Network (IGGCN) for epileptic seizure detection with an average F1-score and recall of 91.5%, and 91.8%, respectively.

Nithya et al. [21] used a majority rule-based Local Binary Pattern (LBP) approach to achieve the highest accuracy of 95.18% on the Freiburg dataset for epileptic seizure detection. Baygin et al. [22] developed a hybrid deep learning network to extract the features of EEG to conduct autism spectrum disorder detection with an accuracy of 96.44%. Kumar et al. [23] extracted both Histogram of Local Variance (HLV) and Symmetrical Weighted Local Binary Pattern (SLBP) features from EEG signals for detecting schizophrenia, realizing an accuracy of 92.85%. Mardini et al. [24] adopted a Genetic Algorithm (GA) in conjunction with four models for EEG signal analysis for epilepsy detection. They found that an Artificial Neural Network (ANN) achieves a higher accuracy. Chen et al. [25] offered a short-time sequence model based on a CNN to extract features from EEG signals for building a detection framework for depression with an accuracy of 99.15%. In other fields, Wang et al. [26] combined time-frequency and non-linear features of EEG to classify bruxism using a fine-tree classifier. Similarly, Bardak et al. [27] proposed a model consisting of Adaptively Designed Neuro-fuzzy Inference System (ANFIS) classifiers in parallel, obtaining great results in emotion recognition utilizing EEG signals.

2.1 REVIEW OF RELATED WORK

Several researchers have explored the use of EEG signals combined with machine learning and deep learning techniques for the detection and classification of neurological and psychiatric disorders. Earlier studies applied signal processing techniques such as Empirical Mode Decomposition (EMD), Wavelet Transform, and Variational Mode Decomposition (VMD) to extract important features from EEG signals before performing classification. These approaches helped improve the accuracy of mental task and disorder identification by analyzing frequency-based characteristics of brain signals.

Many researchers have also used machine learning algorithms such as Support Vector Machines (SVM), Random Forest, and Artificial Neural Networks (ANN) for EEG-based classification tasks. These models demonstrated promising results in detecting disorders such as depression, schizophrenia, and epilepsy by identifying hidden patterns within EEG feature data. Additionally, studies have applied convolutional neural networks (CNN) and recurrent neural networks such as Long Short-Term Memory (LSTM) networks to capture spatial and temporal patterns in EEG signals, improving classification performance.

Recent works have further explored hybrid and advanced models that combine different machine learning techniques to enhance prediction accuracy. Some studies have also used graph-based neural networks and deep learning frameworks to analyze connectivity relationships between EEG signals. These approaches provide better representation of brain activity patterns and help achieve higher classification accuracy for complex neurological and psychiatric disorders. These developments highlight the growing importance of intelligent systems for automated EEG analysis and motivate the development of advanced predictive models in psychiatric disorder identification.

2.2 DEFINITION OF PROBLEM STATEMENT

The primary goal of this project is to develop a robust and scalable deep learning-based system capable of accurately detecting and classifying inappropriate content within YouTube videos. This involves identifying harmful content, categorizing flagged segments, overcoming data limitations, handling real-time scenarios, and mitigating ethical concerns. By addressing these challenges, the system aims to improve user safety.

Further research in EEG-based disorder detection has focused on improving classification

have been widely used to analyze EEG signal images such as scalograms and spectrograms, allowing the model to automatically learn spatial patterns in brain activity. Similarly, Bidirectional Long Short-Term Memory (BiLSTM) networks have been applied to capture temporal dependencies in EEG time-series data, enabling better recognition of mental states and emotional patterns. These approaches have shown promising results in emotion recognition, mental task classification, and neurological disorder detection.

Researchers have also explored the use of ensemble learning techniques, where multiple classifiers are combined to improve prediction accuracy and reliability. Methods such as Random Forest and boosting algorithms have demonstrated strong performance in handling high-dimensional EEG feature sets and reducing overfitting problems. Additionally, feature extraction techniques such as Local Binary Pattern (LBP), Discrete Wavelet Transform (DWT), and statistical feature analysis have been used to improve the representation of EEG signals before classification.

More recent studies have focused on graph-based neural networks and hybrid learning frameworks to better model the complex relationships between EEG channels and brain regions. These models treat EEG features as interconnected nodes in a graph structure, enabling the system to capture spatial dependencies and functional connectivity within brain signals.

2.3 EXISTING SYSTEM

In the traditional healthcare framework, the diagnosis and classification of neurological or psychological disorders using EEG data were primarily carried out through manual analysis and conventional statistical techniques. Medical professionals relied heavily on visual interpretation of EEG waveforms, clinical observations, and patient history to determine abnormalities. Earlier computational approaches used basic signal processing and traditional machine learning algorithms such as K-Nearest Neighbors (KNN), Naive Bayes, or Linear Discriminant Analysis (LDA) to classify disorders. Although these methods provided initial insights, they lacked the capability to capture the complex, nonlinear patterns inherent in EEG signals.

Moreover, traditional systems required extensive manual feature extraction, which made the process time-consuming and prone to human error. These models were limited in scalability and accuracy due to their dependence on handcrafted features and the inability to learn deep representations of data. Additionally, traditional systems lacked integration with automated prediction or real-time analysis tools, making them unsuitable for large-scale applications.

Limitations of Existing System

- Diagnosis and feature selection rely heavily on expert knowledge, leading to subjective and inconsistent results.
- Manual EEG signal analysis and feature extraction are time-consuming and not suitable for real-time applications.
- Traditional methods fail to capture complex, nonlinear, and spatial relationships in EEG signals.
- The system struggles with scalability, making it inefficient at handling large EEG datasets or high-dimensional features.
- Classical models like KNN, Naïve Bayes, or LDA often result in lower classification performance compared to modern deep learning approaches.
- No integrated pipeline exists for preprocessing, training, and prediction, with most steps performed separately.
- Manual data interpretation increases the chances of diagnostic errors.
- Traditional models have poor generalization and struggle to adapt to unseen or noisy EEG data from different patients.
- The system cannot provide immediate or continuous diagnostic insights during EEG recording.

2.4 PROPOSED SYSTEM

The proposed system presents an intelligent EEG-based disorder classification model that integrates advanced Machine Learning (ML) and Hybrid Graph Neural Network (HGNN) algorithms to overcome the limitations of traditional diagnostic methods. Unlike conventional systems that depend heavily on manual feature extraction and expert judgment, this system automates the entire workflow from data preprocessing to real-time prediction through a unified, Flask-based web platform. The system is capable of learning complex, nonlinear relationships in EEG data, identifying both main disorders and specific disorders with high accuracy. The process begins with the collection of EEG signal datasets, which undergo a detailed preprocessing pipeline involving noise removal, normalization, and label encoding to prepare them for training. The cleaned dataset is then used to train multiple classification models such as Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), Decision Tree Classifier (DTC), and the Hybrid Graph Neural Network (HGNN). Each model's performance is evaluated and compared using key performance indicators including accuracy, precision, recall, and F1-score.

Advantages of the Proposed System:

The proposed system significantly improves upon the existing approaches by addressing key limitations:

- evaluations and minimizing diagnostic bias in psychiatric disorder identification.
- Handles complex and high-dimensional EEG data effectively, capturing nonlinear relationships and hidden patterns that are difficult to detect using traditional analysis methods.
- Supports accurate multi-class and multi-level classification, enabling both generalized and fine-grained identification of psychiatric disorders.
- Offers comparative model evaluation, allowing selection of the most suitable classifier based on performance metrics for real-world deployment.
- Ensures fast and automated prediction, improving efficiency in clinical workflows and reducing diagnosis time.
- Improves reliability and consistency of diagnostic outcomes, leading to better patient management and treatment planning.

2.5 OBJECTIVES

▪ **Develop an Intelligent EEG-Based Disorder Detection System** – Design a machine learning–based system capable of analyzing EEG signal features to automatically identify psychiatric and neurological disorders.

▪ **Automate EEG Data Analysis** – Reduce dependence on manual EEG interpretation by implementing automated preprocessing, feature extraction, and classification techniques.

▪ **Improve Diagnostic Accuracy** – Apply multiple machine learning algorithms such as Random Forest, Support Vector Machine, Logistic Regression, Decision Tree, and Hybrid Graph Neural Network to accurately classify psychiatric disorders.

▪ **Perform Multi-Level Disorder Classification** – Enable the system to classify both main disorder categories and specific disorder types using EEG feature data.

▪ **Provide a Web-Based Prediction Platform** – Develop a Flask-based web application that allows administrators to train and evaluate models and enables users to upload EEG datasets for real-time disorder prediction.

▪ **Support Model Evaluation and Comparison** – Implement performance evaluation using metrics such as accuracy, precision, recall, and F1-score to identify the most effective classification model.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS:

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- Processor : Intel Core i5
- Memory : 1TB.
- RAM : 8GB.

2.6.2 SOFTWARE REQUIREMENTS:

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

- Operating system : Windows
- Language : Python 3.12.0
- Back-End : Flask Server
- Frontend : HTML,CSS
- Database : SQL Lite

3.SYSTEM ARCHITECTURE & DESIGN

3.SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, enhancing collaboration and reducing complexity.

3.1 PROJECT ARCHITECTURE

This project is a Flask-based web application designed for automatic classification and analysis of EEG data using multiple machine learning algorithms. It supports secure user management with role-based access, offering specialized functionality for both admin and regular users.

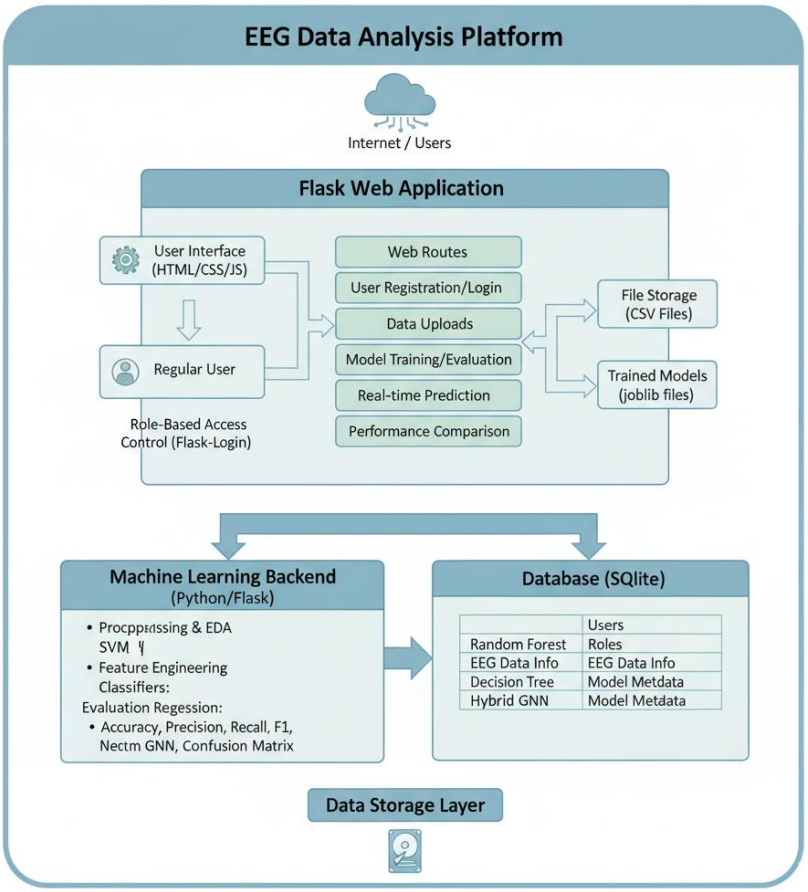

Figure 3.1: System architecture of proposed psychiatric disorder identification

3.2 DESCRIPTION

Input Data:

The project uses EEG (Electroencephalography) signal data containing numerical brain activity features recorded from multiple electrodes. The dataset includes EEG-derived features along with two target labels: main disorder and specific disorder, representing different psychiatric conditions.

Reading Data:

The EEG dataset is loaded using the Pandas library in Python. The system reads CSV files containing EEG features and performs an initial inspection to identify columns, feature values, and target labels before processing.

Data Preprocessing:

The raw EEG data undergoes preprocessing steps including removal of irrelevant columns, handling missing values using mean imputation, noise filtering, and normalization to ensure clean and consistent input for machine learning models.

Feature Extraction:

Important EEG features such as absolute band power values across different frequency bands (delta, theta, alpha, beta, and gamma) are extracted and used as inputs for classification models.

Model Training:

Multiple machine learning models including Random Forest, Support Vector Machine (SVM), Logistic Regression, Decision Tree, and Hybrid Graph Neural Network (HGNN) are trained on the processed EEG dataset to learn patterns associated with psychiatric disorders.

Classification Layer:

The trained models classify EEG signals into disorder categories by predicting both main disorder types and specific disorder classes based on extracted EEG features.

Training and Evaluation:

The models are evaluated using performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix to compare their effectiveness in detecting psychiatric disorders.

Prediction and Feedback:

Users can upload new EEG datasets through the Flask-based web interface, and the system generates real-time predictions. The results help researchers and clinicians analyze brain activity patterns and support psychiatric disorder diagnosis.

3.3 UML Diagrams

Unified Modeling Language (UML) is a standardized visual language used to model, design, and document the architecture of software systems. It provides a set of graphical notations to represent the structure and behavior of a system, making complex systems easier to understand and communicate among developers, stakeholders, and business analysts.A Data Flow Diagram mainly consists of **four important components**:

3.3.1 Class Diagram

The class diagram illustrates the structural design of the system, showing relationships between user management, Flask-based application logic, data preprocessing, machine learning classifiers, and model storage. It captures inheritance among classifiers and highlights how EEG data flows through preprocessing, training, and prediction modules

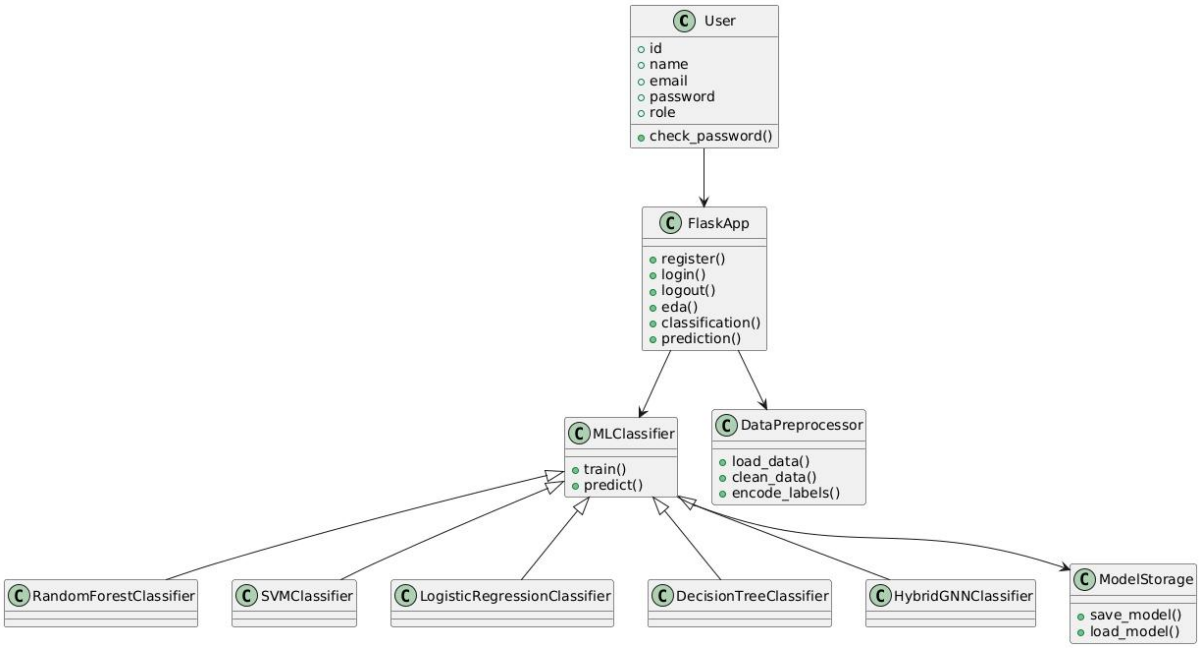

Fig. 3.3.1 : Class Diagram of Neural network based predictive modeling for complex psychiatric disorder

3.3.2 Use Case Diagram

This use case diagram explicitly shows how both admin and user interact with each individual algorithm. Admin users train and evaluate each classifier independently, while normal users obtain predictions from all trained models, ensuring complete transparency and algorithm-level interaction.

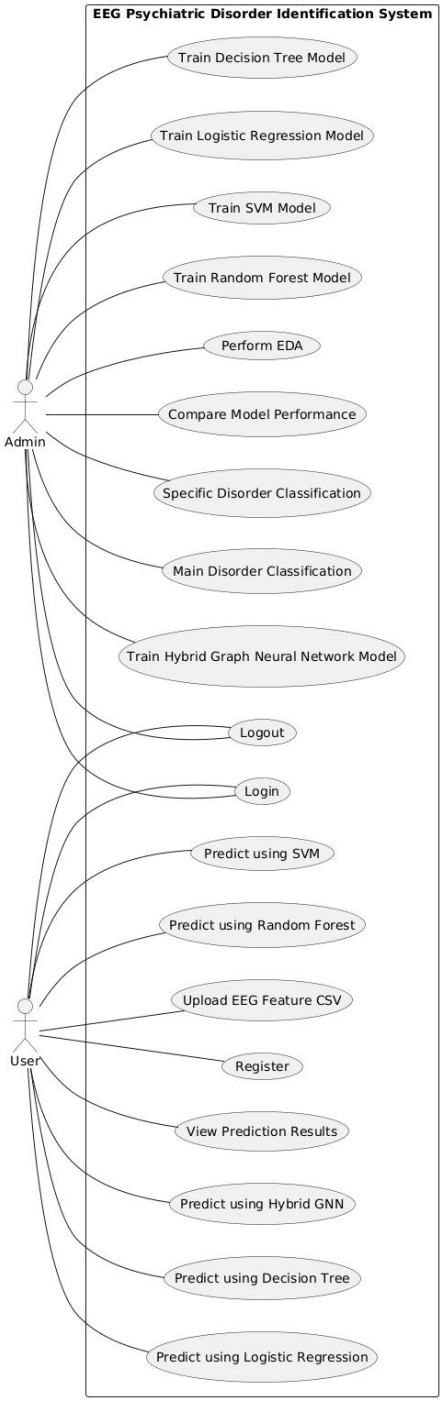

Fig. 3.3.2: Use Case Diagram.Class Diagram of Neural network based predictive modeling for complex psychiatric disorder

3.3.3 Activity Diagram

This activity diagram demonstrates the complete operational workflow, clearly separating admin-based training and user-based prediction while explicitly including every algorithm at both stages.

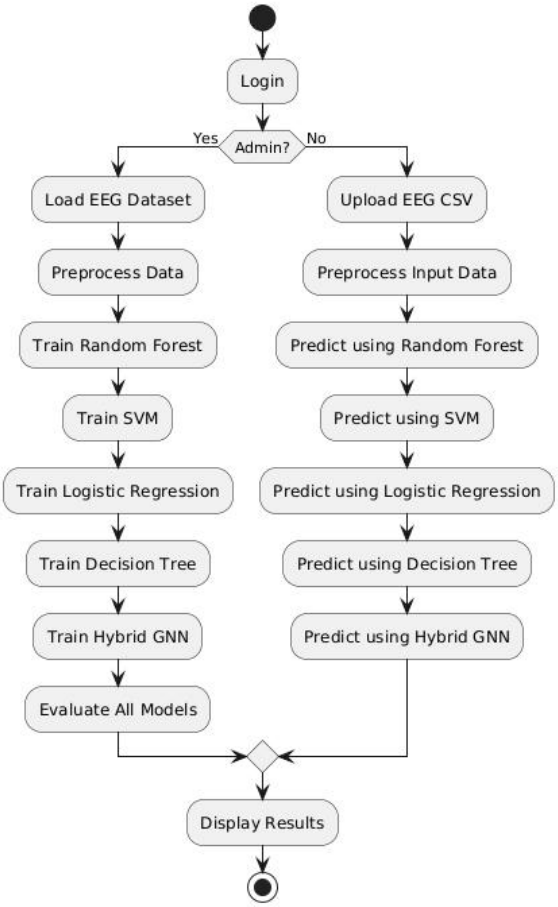

Fig. 3.3.3: Activity Diagram.

3.3.4 Sequence Diagram

The sequence diagram clearly shows how EEG data flows through every trained algorithm individually during prediction. Each classifier independently generates predictions, which are aggregated and returned to the user, ensuring comprehensive comparative analysis.

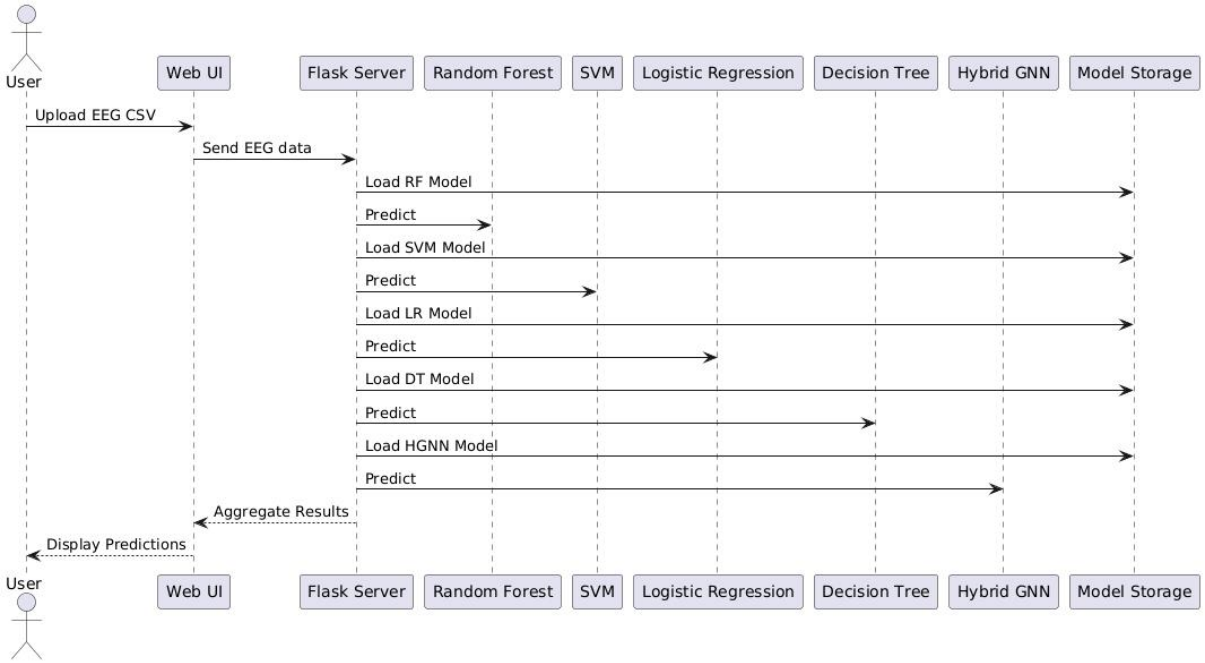

Fig. 3.3.4: Sequence Diagram.Class Diagram of Neural network based predictive modeling for complex psychiatric disorder

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of a project involves executing the planned strategies and tasks. It requires meticulous coordination, resource allocation, and monitoring to ensure that objectives are met efficiently. Effective implementation is crucial for achieving project goals and delivering expected outcomes within the set timeline and budget constraints.

4.1 ALGORITHMS USED

In the proposed EEG-based psychiatric disorder identification system, several machine learning algorithms are used to analyze brain signal features and classify neurological and psychiatric disorders. EEG signals contain complex and high-dimensional data representing electrical activity of the brain across different frequency bands such as delta, theta, alpha, beta, and gamma. These signals often contain nonlinear relationships and hidden patterns that are difficult to interpret using traditional analysis methods. To effectively capture these patterns, the system employs multiple classification algorithms including Random Forest, Support Vector Machine (SVM), Logistic Regression, Decision Tree, and a Hybrid Graph Neural Network (HGNN). These algorithms are trained on preprocessed EEG datasets to learn relationships between EEG features and disorder labels, enabling the system to predict both main disorders and specific psychiatric conditions. Using multiple algorithms also allows the system to compare performance metrics and identify the most accurate model for real-world clinical applications.

Random Forest (RF) for Robust Classification

Random Forest is an ensemble machine learning algorithm that constructs multiple decision trees during the training phase and combines their outputs to produce a final prediction. Each decision tree is trained using a randomly selected subset of EEG data and features. This randomness increases model diversity and reduces overfitting.

In the context of EEG disorder classification, Random Forest analyzes multiple brain signal features simultaneously and determines patterns that correspond to different psychiatric conditions. Each tree independently predicts a disorder class, and the final prediction is obtained through majority voting.

Advantages of Random Forest

• Handles high-dimensional EEG datasets effectively.
• Captures nonlinear relationships between EEG features.
• Reduces overfitting through ensemble learning.
• Provides strong classification performance for complex datasets.

Disadvantages of Random Forest

• Requires higher memory due to multiple decision trees.
• Prediction time increases with a large number of trees.
• Less interpretable compared to simple models.

Support Vector Machine (SVM) for High-Dimensional Data Analysis

Support Vector Machine (SVM) is a powerful supervised learning algorithm widely used for classification problems involving high-dimensional data. The SVM algorithm works by identifying an optimal hyperplane that separates EEG feature vectors belonging to different psychiatric disorder classes.

In this project, EEG signal features are represented as numerical vectors, and SVM analyzes these vectors to determine the boundary between different disorder categories. When EEG data is not linearly separable, kernel functions such as radial basis function (RBF) can be applied to transform the data into a higher-dimensional space where separation becomes possible.

Advantages of SVM

• Performs well in high-dimensional EEG feature spaces.
• Provides strong generalization capability.
• Effective for complex classification tasks.

Disadvantages of SVM

• Training time increases for large EEG datasets.
• Requires careful parameter tuning for optimal performance.

Logistic Regression for Baseline Classification

Logistic Regression is a statistical machine learning algorithm used for classification tasks. It estimates the probability of an EEG sample belonging to a particular psychiatric disorder class by applying a sigmoid function to a weighted combination of input features.

In this system, Logistic Regression serves as a baseline classifier that helps evaluate the performance of more complex algorithms. It predicts disorder classes based on EEG feature values such as band power measurements and connectivity patterns.

Advantages of Logistic Regression

• Simple and easy to implement.
• Computationally efficient and fast.
• Provides probabilistic outputs for classification.

Disadvantages of Logistic Regression

• Assumes linear relationships between EEG features and disorder classes.
• Less effective for complex nonlinear patterns in brain signals.

Decision Tree Classifier for Interpretable Predictions

Decision Tree is a tree-structured classification algorithm that divides the EEG dataset into smaller subsets based on feature thresholds. Each internal node represents a decision rule based on EEG feature values, while leaf nodes represent final disorder classes.

The algorithm recursively splits the dataset until the most informative features are identified for classification. This hierarchical structure allows researchers and clinicians to understand how specific EEG features influence the prediction outcome.

Advantages of Decision Tree

• Easy to interpret and visualize.
• Clearly shows the decision rules used for classification.
• Requires minimal data preprocessing.

Disadvantages of Decision Tree

• Prone to overfitting when the tree becomes too deep.
• Less stable compared to ensemble methods.

Hybrid Graph Neural Network (HGNN) for Advanced EEG Analysis

The Hybrid Graph Neural Network (HGNN) is an advanced model designed to capture complex relationships between EEG features. Unlike traditional algorithms that treat features independently, HGNN models EEG signals as interconnected nodes in a graph structure.

In this representation, each EEG feature (such as band power or connectivity measurement) is treated as a node, and relationships between features are represented as edges. The model aggregates information from neighboring nodes, allowing it to capture spatial and functional relationships between brain regions.

This graph-based learning approach enables the system to identify hidden dependencies in EEG signals, improving the accuracy of psychiatric disorder classification.

Advantages of HGNN

• Captures complex nonlinear relationships in EEG data.
• Models connectivity patterns between EEG features.
• Provides higher accuracy for disorder classification.
• Effective for both main and specific disorder prediction.

Disadvantages of HGNN

• Requires more computational resources.
• More complex to implement compared to classical algorithms.

4.2 SAMPLE CODE

```
from flask import Flask, render_template, request, redirect, url_for, flash, jsonify, session
from flask_login import LoginManager, login_user, logout_user, login_required, current_user
from werkzeug.utils import secure_filename
import os
import pandas as pd
from functools import wraps
from models import init_db, User
from ml_classifiers import train_all_classifiers, predict_from_csv, get_eda_stats
app = Flask(__name__)
app.config['SECRET_KEY'] = os.environ.get('SESSION_SECRET', 'your-secret-key-here')
app.config['UPLOAD_FOLDER'] = 'uploads'
app.config['MAX_CONTENT_LENGTH'] = 16 * 1024 * 1024

os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)
os.makedirs('static/plots', exist_ok=True)

login_manager = LoginManager()
login_manager.init_app(app)
login_manager.login_view = 'login'
@login_manager.user_loader
def load_user(user_id):
    return User.get(int(user_id))
def admin_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if not current_user.is_authenticated or current_user.role != 'admin':
            flash('You need admin privileges to access this page.', 'danger')
            return redirect(url_for('home'))
        return f(*args, **kwargs)
    return decorated_function

def user_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if not current_user.is_authenticated or current_user.role != 'user':
```

```
            flash('This page is only accessible to regular users.', 'danger')
            return redirect(url_for('home'))
        return f(*args, **kwargs)
    return decorated_function
@app.route('/')
def index():
    return redirect(url_for('home'))

@app.route('/home')
@login_required
def home():
    return render_template('home.html')

@app.route('/register', methods=['GET', 'POST'])
def register():
    if request.method == 'POST':
        name = request.form.get('name')
        mobile = request.form.get('mobile')
        email = request.form.get('email')
        address = request.form.get('address')
        password = request.form.get('password')
        confirm_password = request.form.get('confirm_password')

        if password != confirm_password:
            flash('Passwords do not match!', 'danger')
            return redirect(url_for('register'))

        if User.create(name, mobile, email, address, password, 'user'):
            flash('Registration successful! Please login.', 'success')
            return redirect(url_for('login'))
        else:
            flash('Email already exists!', 'danger')
            return redirect(url_for('register'))

    return render_template('register.html')

@app.route('/login', methods=['GET', 'POST'])
def login():
```

```
    if request.method == 'POST':
        email = request.form.get('email')
        password = request.form.get('password')
        user = User.get_by_email(email)

        if user and user.check_password(password):
            login_user(user)
            flash('Login successful!', 'success')
            return redirect(url_for('home'))
        else:
            flash('Invalid email or password!', 'danger')
            return redirect(url_for('login'))

    return render_template('login.html')

@app.route('/logout')
@login_required
def logout():
    logout_user()
    flash('You have been logged out.', 'info')
    return redirect(url_for('login'))

@app.route('/eda')
@login_required
@admin_required
def eda():
    stats = get_eda_stats()
    return render_template('eda.html', stats=stats)

# ---------------- Main Disorder Classification ----------------
@app.route('/main_disorder_classification', methods=['GET', 'POST'])
@login_required
@admin_required
def main_disorder_classification():
    classifiers = [
        'Random Forest',
        'Support Vector Machine',
        'Logistic Regression',
```

```
        'Decision Tree',
        'Hybrid Graph Neural Network'
    ]

    if request.method == 'POST':
        selected_classifiers = request.form.getlist('classifiers')

        if not selected_classifiers:
            flash('Please select at least one classifier!', 'warning')
            return redirect(url_for('main_disorder_classification'))

        results = train_all_classifiers('main.disorder')
        filtered_results = {k: v for k, v in results.items() if k in selected_classifiers}

        return render_template(
            'classification_results.html',
            results=filtered_results,
            target='Main Disorder',
            target_column='main.disorder',
            endpoint='main_disorder_classification'  # Pass endpoint for Jinja2
        )

    return render_template(
        'classifier_selection.html',
        classifiers=classifiers,
        target='Main Disorder',
        endpoint='main_disorder_classification'
    )

# ---------------- Specific Disorder Classification ----------------
@app.route('/specific_disorder_classification', methods=['GET', 'POST'])
@login_required
@admin_required
def specific_disorder_classification():
    classifiers = [
        'Random Forest',
        'Support Vector Machine',
        'Logistic Regression',
```

```
        'Decision Tree',
        'Hybrid Graph Neural Network'
    ]

    if request.method == 'POST':
        selected_classifiers = request.form.getlist('classifiers')

        if not selected_classifiers:
            flash('Please select at least one classifier!', 'warning')
            return redirect(url_for('specific_disorder_classification'))

        results = train_all_classifiers('specific.disorder')
        filtered_results = {k: v for k, v in results.items() if k in selected_classifiers}

        return render_template(
            'classification_results.html',
            results=filtered_results,
            target='Specific Disorder',
            target_column='specific.disorder',
            endpoint='specific_disorder_classification'  # Pass endpoint
        )

    return render_template(
        'classifier_selection.html',
        classifiers=classifiers,
        target='Specific Disorder',
        endpoint='specific_disorder_classification'
    )

# ---------------- Performance Comparison ----------------
@app.route('/performance_comparison')
@login_required
@admin_required
def performance_comparison():
    main_results = {}
    specific_results = {}

    if os.path.exists('model'):
```

```
        main_results = train_all_classifiers('main.disorder')
        specific_results = train_all_classifiers('specific.disorder')

    return render_template(
        'performance_comparison.html',
        main_results=main_results,
        specific_results=specific_results
    )

# ---------------- User Prediction ----------------
@app.route('/prediction', methods=['GET', 'POST'])
@login_required
@user_required
def prediction():
    if request.method == 'POST':
        if 'file' not in request.files:
            flash('No file uploaded!', 'danger')
            return redirect(url_for('prediction'))

        file = request.files['file']
        target_column = request.form.get('target_column')

        if file.filename == '':
            flash('No file selected!', 'danger')
            return redirect(url_for('prediction'))

        if file and file.filename.endswith('.csv'):
            filename = secure_filename(file.filename)
            filepath = os.path.join(app.config['UPLOAD_FOLDER'], filename)
            file.save(filepath)
            try:
                predictions = predict_from_csv(filepath, target_column)
                df_original = pd.read_csv(filepath)
                return render_template(
                    'prediction_results.html',
                    predictions=predictions,
                    target=target_column,
                    num_samples=len(df_original)
```

```
            )
        except Exception as e:
            flash(f'Error processing file: {str(e)}', 'danger')
            return redirect(url_for('prediction'))
    else:
        flash('Please upload a CSV file!', 'danger')
        return redirect(url_for('prediction'))

    return render_template('prediction.html')
# ---------------- Main ----------------
if __name__ == '__main__':
    init_db()
    app.run(host='0.0.0.0', port=5000, debug=True)
        main.mainloop()
```

5. RESULTS & DISCUSSION

5. RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

5.1 System Home Interface :

The home page introduces the EEG Psychiatric Disorder Classification System. It explains that the application uses machine learning techniques to analyze EEG features for psychiatric disorder identification. The interface also displays the available classification models and system features.

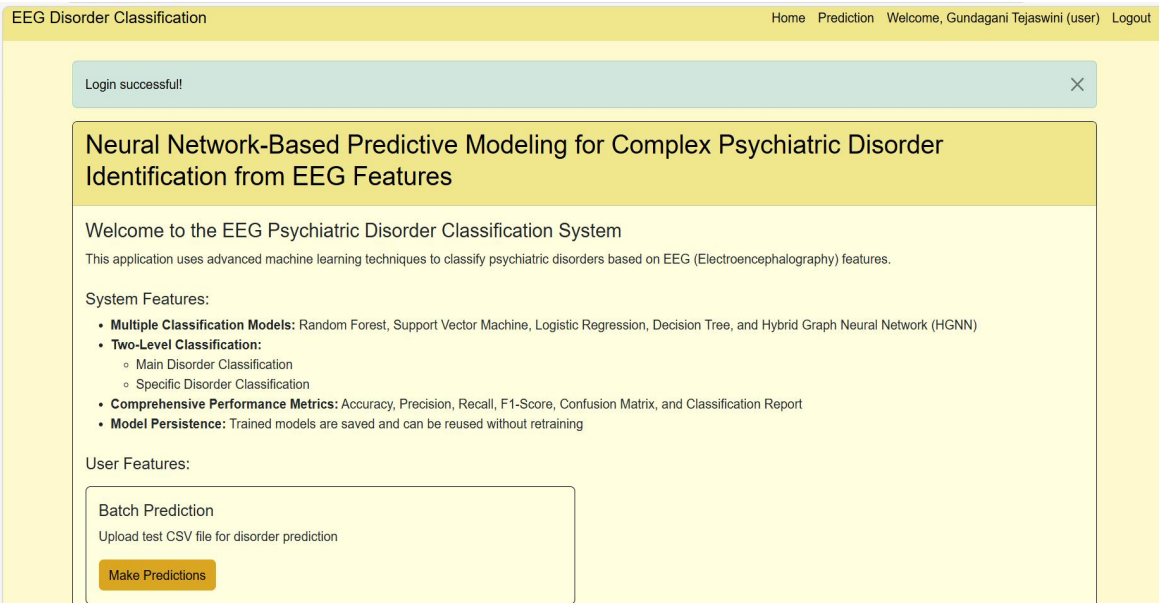

Figure 5.1 : System home interface of Neural Network-Based Predictive Modeling for Complex Psychiatric Disorder Identification using EEG Features

5.2 Classifier Selection interface :

In this screen, the administrator selects one or more machine learning classifiers to train and analyze. The available classifiers include Random Forest, Support Vector Machine, Logistic Regression, Decision Tree, and Hybrid Graph Neural Network. After selection, the system trains the chosen models using the EEG dataset.

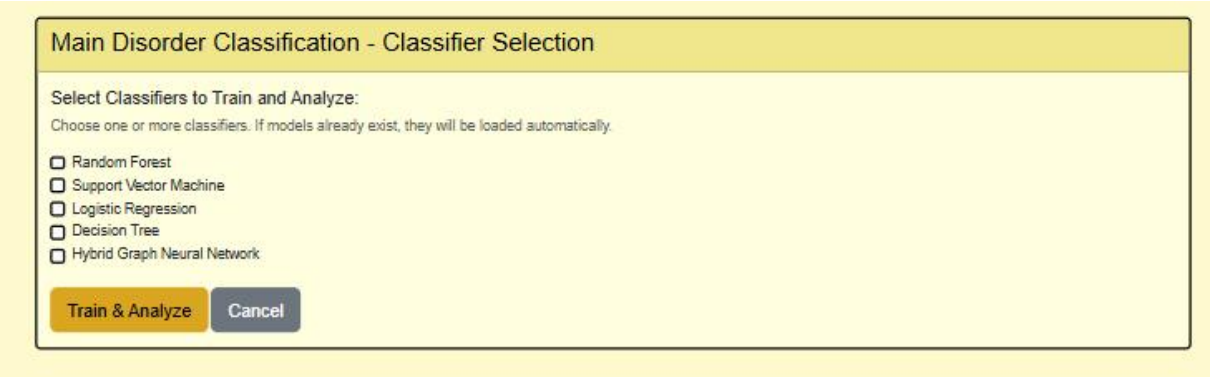

Figure 5.2 : Classifier selection interface of the Neural Network-Based Predictive Modeling for Complex Psychiatric Disorder Identification using EEG Features

5.3 Main Disorder Classification :

This section shows the classification of EEG data into major psychiatric disorder categories. The system processes EEG features and predicts disorders such as anxiety disorder, mood disorder, schizophrenia, and trauma-related disorders. The results are generated using the trained machine learning models

Sample #	Predicted main.disorder
1	Addictive disorder
2	Addictive disorder
3	Addictive disorder
4	Anxiety disorder
5	Anxiety disorder
6	Anxiety disorder
7	Anxiety disorder
8	Healthy control
9	Healthy control
10	Healthy control
11	Healthy control
12	Healthy control
13	Mood disorder
14	Mood disorder
15	Mood disorder
16	Obsessive compulsive disorder

Figure 5.3 : Main disorder classification of the Neural Network-Based Predictive Modeling for Complex psychiatric disorder identification using EEG features

5.4 Specific Disorder Classification :

In this stage, the system performs a detailed classification of EEG signals into specific psychiatric disorders. The models analyze EEG patterns to provide fine-grained predictions beyond the main disorder categories. This helps improve diagnostic accuracy.

Sample #	Predicted specific.disorder
1	Depressive disorder
2	Alcohol use disorder
3	Alcohol use disorder
4	Depressive disorder
5	Alcohol use disorder
6	Depressive disorder
7	Social anxiety disorder
8	Depressive disorder
9	Depressive disorder
10	Depressive disorder
11	Healthy control
12	Depressive disorder
13	Depressive disorder
14	Bipolar disorder
15	Depressive disorder
16	Depressive disorder
17	Depressive disorder
18	Depressive disorder

Figure 5.4 : Specific disorder classification of the Neural Network-Based Predictive Modeling for Complex psychiatric disorder identification using EEG features

5.5 Performance Comparision of Classifiers:

The system compares the performance of different classifiers using evaluation metrics such as accuracy. The results show how each model performs in classifying EEG disorders. This comparison helps determine the most effective algorithm.

Prediction Distribution

Disorder	Count	Percentage
Addictive disorder	3	11.11%
Anxiety disorder	4	14.81%
Healthy control	5	18.52%
Mood disorder	3	11.11%
Obsessive compulsive disorder	4	14.81%
Schizophrenia	3	11.11%
Trauma and stress related disorder	5	18.52%

Figure 5.5 : Performance Comparision of Classifiers Neural Network-Based Predictive Modeling for Complex Psychiatric Disorder Identification using EEG Features

5.6 Evaluation Metrics and Classification Report :

The system generates evaluation metrics including precision, recall, F1-score, and accuracy. A classification report is displayed to show the performance of each disorder category. These metrics help analyze the reliability of the trained models.

Main Disorder Classification

Classifier	Accuracy	Precision	Recall	F1-Score
Random Forest	0.5437	0.7473	0.5437	0.5066
Support Vector Machine	0.6689	0.6803	0.6689	0.6652
Logistic Regression	0.5917	0.6036	0.5917	0.5855
Decision Tree	0.3558	0.2640	0.3558	0.2571
Hybrid Graph Neural Network	1.0000	1.0000	1.0000	1.0000

Specific Disorder Classification

Classifier	Accuracy	Precision	Recall	F1-Score
Random Forest	0.4021	0.7831	0.4021	0.3627
Support Vector Machine	0.3241	0.3368	0.3241	0.3178
Logistic Regression	0.5904	0.6057	0.5904	0.5813
Decision Tree	0.2416	0.1843	0.2416	0.1333
Hybrid Graph Neural Network	1.0000	1.0000	1.0000	1.0000

Figure 5.6 : Evaluation Metrics and Classification Report of Neural Network-Based Predictive Modeling for Complex Psychiatric Disorder Identification using EEG Features

5.7 Final prediction Output:

The trained models are used to predict psychiatric disorders based on EEG features. The prediction results are displayed in the system interface along with classification details. This allows clinicians or researchers to interpret the diagnostic results easily.

Disorder	Count	Percentage
Addictive disorder	3	11.11%
Anxiety disorder	4	14.81%
Healthy control	5	18.52%
Mood disorder	3	11.11%
Obsessive compulsive disorder	4	14.81%
Schizophrenia	3	11.11%
Trauma and stress related disorder	5	18.52%

Figure 5.7 : Final prediction output of Neural Network-Based Predictive Modeling for Complex Psychiatric Disorder Identification using EEG Features

6.VALIDATION

6. VALIDATION

The validation of this project is performed through systematic testing and evaluation of the EEG-based psychiatric disorder classification system. The validation process ensures that the proposed machine learning models can accurately identify psychiatric disorders using EEG features. The system is tested at different stages, including dataset validation, preprocessing verification, model training evaluation, and prediction testing. Through this structured validation process, the system's reliability and accuracy are verified while minimizing classification errors.

6.1 INTRODUCTION

During the validation phase, the EEG dataset is divided into **training and testing datasets**, usually using an **80:20 split ratio**. The training dataset is used to train the machine learning models, while the testing dataset is used to evaluate how well the models perform on previously unseen EEG data. This process helps determine the model's ability to generalize to new data.

To further enhance reliability, additional validation techniques such as **cross-validation** can be applied. Cross-validation allows the dataset to be tested across multiple partitions, ensuring that the model does not overfit the training data. This approach improves the robustness and stability of the classification models.

Finally, the performance of all machine learning models is compared to identify the most effective classifier for EEG-based psychiatric disorder detection. This validation process confirms that the proposed system is reliable and capable of assisting researchers and clinicians in analyzing EEG signals for psychiatric disorder identification.

6.2 TEST CASES

TABLE 6.2.1 UPLOADING EEG DATASET

Test case ID	Test case name	Purpose	Test Case	Output
1	User uploads EEG Dataset	To verify that the system accepts and loads EEG dataset correctly	The user uploads an EEG dataset file(CSV) through the system interface	Dataset successfully loaded and ready for preprocessing

TABLE 6.2.2 CLASSIFICATION

Test case ID	Test case name	Purpose	Input	Output
1	Classification test 1	To verify that the classifier predicts the main disorder correctly	EEG feature dataset is selected	Predicted main disorder displayed
2	Classification test 2	To verify that the classifier predicts the specific disorder correctly	EEG feature dataset is selected	Predicted specific disorder displayedt

7. CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, the proposed EEG-based psychiatric disorder identification system successfully demonstrates the potential of machine learning techniques in analyzing brain signal data for automated diagnosis support. The system integrates multiple machine learning algorithms with a Flask-based web application to provide an intelligent platform for analyzing EEG features and predicting psychiatric disorders. Through systematic preprocessing, feature analysis, and model training, the system is capable of identifying both main and specific psychiatric disorders from EEG data with reliable accuracy.

7.1 PROJECT CONCLUSION

This project presents an intelligent machine learning-based framework for identifying psychiatric disorders using EEG signal features. The system utilizes several classification algorithms, including **Random Forest, Support Vector Machine (SVM), Logistic Regression, Decision Tree, and Hybrid Graph Neural Network (HGNN)**, to analyze complex EEG patterns and classify disorders accurately. These models are trained on preprocessed EEG datasets and evaluated using performance metrics such as **accuracy, precision, recall, F1-score, and confusion matrix analysis**.

The integration of a **Flask-based web application** enables users to upload EEG feature datasets and obtain real-time disorder predictions. The system also provides exploratory data analysis, model training, and performance comparison functionalities. Among the implemented models, the Hybrid Graph Neural Network demonstrates improved performance in capturing nonlinear relationships between EEG features.

By combining machine learning techniques with an interactive web interface, the proposed system provides a scalable and efficient solution for EEG-based psychiatric disorder identification. The results indicate that such intelligent systems can support clinicians and researchers in early detection and analysis of neurological and psychiatric conditions.

7.2 FUTURE ASPECTS

Although the proposed system demonstrates promising results, several improvements can be made in the future to enhance its functionality and real-world applicability. Future work may focus on integrating advanced deep learning architectures for improved EEG signal analysis and higher prediction accuracy.

Some potential future enhancements include:

• **Integration of Deep Learning Models:** Implementing advanced neural network models such as Convolutional Neural Networks (CNN) or Graph Neural Networks for improved EEG feature learning.
• **Real-Time EEG Monitoring:** Extending the system to support real-time EEG signal processing for continuous psychiatric monitoring.
• **Larger and Diverse Datasets:** Expanding the dataset with more EEG recordings from diverse populations to improve model generalization.
• **Clinical Decision Support Integration:** Integrating the system with hospital information systems to assist clinicians in diagnosis and treatment planning.
• **Improved Visualization Tools:** Developing advanced visualization dashboards for better interpretation of EEG patterns and model predictions.
• **Cloud-Based Deployment:** Deploying the system on cloud platforms to support large-scale data processing and remote accessibility.

With these improvements, the system can evolve into a more powerful tool for assisting mental health professionals and advancing research in EEG-based psychiatric disorder detection.

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8. BIBLIOGRAPHY

8. BIBLIOGRAPHY

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8.2 GITHUB LINK

https://github.com/gundaganitejaswini05/Neural-Network-Based-Predictive-Modeling-for-Complex-Psychiatric-Disorders